Dr Augustinas Silale

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Education

2015-2019	Wellcome Trust Cellular Structural Biology DPhil Programme, University of Oxford, UK. Thesis: "Investigation of the DNA uptake machinery in naturally competent bacteria." Supervised by Prof Ben Berks and Prof Susan Lea.
2011-2015	BA & MSci Natural Sciences (Biochemistry), 1st Class (Hons), University of Cambridge, UK. BA Project with Prof Sir Tom Blundell, MSci Project with Dr Ingo Greger.
2011	Brandos Atestatas (*cum laude*), Panevezys Juozas Balcikonis Gymnasium, Lithuania.

Employment

01/2025-present	Postdoctoral Research Associate, Tracy Palmer & Lis Lowe groups, Newcastle, UK.
11/2019-12/2024	Postdoctoral Research Associate, Bert van den Berg group, Newcastle University, UK.
06/2015-08/2015	Research Support Assistant, Ingo Greger group, MRC LMB, Cambridge, UK.
06/2014-08/2015	Summer Internship, Tom Blundell group, University of Cambridge, UK.
06/2013-09/2015	Research Support Assistant, Rob Kay group, MRC LMB, Cambridge, UK.
06/2012-08/2012	Research Support Assistant, Leo James group, MRC LMB, Cambridge, UK.

Research outputs

Preprint:

Silale A[#], Soo YL, Mark H, Baslé A, van den Berg B[#] (2024). Structural basis of iron piracy by a prominent human gut symbiont. *bioRxiv*. [#]**Co-corresponding author**.

Peer-reviewed publications:

Silale A, Zhu Y, Witwinowski J, Smith RE, Newman KE, Bhamidimarri SP, Baslé A, Khalid S, Beloin C, Gribaldo S, van den Berg B (2023) Dual function of OmpM as outer membrane tether and nutrient uptake channel in diderm Firmicutes. *Nat Commun* **14**, 7152.

Abellon-Ruiz J, Jana K, **Silale A**, Frey AM, Baslé A, Trost M, Kleinekathöfer U, van den Berg B (2023) BtuB TonB-dependent transporters and BtuG surface lipoproteins form stable complexes for vitamin B_{12} uptake in gut *Bacteroides*. *Nat Commun* **14**, 4714.

White JBR*, **Silale A***, Feasey M, Heunis T, Zhu Y, Zheng H, Gajbhiye A, Firbank S, Baslé A, Trost M, Bolam DN, van den Berg B, Ranson NA (2023) Outer membrane utilisomes mediate glycan uptake in gut Bacteroidetes. *Nature* **618**, 583–589. ***Co-first author**.

van den Berg B, **Silale A**, Baslé A, Brandner AF, Mader SL, Khalid S (2022) Structural basis for host recognition and superinfection exclusion by bacteriophage T5. *Proc Natl Acad Sci USA*. 119(42):e2211672119.

Silale A, Lea SM, Berks BC (2021) The DNA transporter ComEC has metal-dependent nuclease activity that is important for natural transformation. *Mol Microbiol* **116**, 416-426.

Hennell James R, Deme JC, Kjær A, Alcock F, **Silale A**, Lauber F, Johnson S, Berks BC, Lea SM (2021) Structure and mechanism of the proton-driven motor that powers type 9 secretion and gliding motility. *Nat Microbiol* **6**, 221–233.

Paschke P, Knecht DA, **Silale A**, Traynor D, Williams TD, Thomason PA, Insall RH, Chubb JR, Kay RR & Veltman DM (2018) Rapid and efficient genetic engineering of both wild type and axenic strains of *Dictyostelium discoideum*. *PLoS One* 13: e0196809.

Review:
Silale A[#], van den Berg B[#] (2023) TonB-Dependent Transport Across the Bacterial Outer Membrane. *Annu Rev Microbiol* **77**, 67–88. [#]**Co-corresponding author.**

Funding

- BBSRC ALERT 2023 Equipment Grant: Northern Eye - A Cryo-Electron Microscopy Resource in the North East, 2024 (Project Co-Lead, £1.4M)
- Microbiology Society Travel Grant, 2019 (Project Lead)
- Wellcome Trust 4-year DPhil Studentship, 2015 (Project Lead, £162K)

Awards

- Best Paper Prize, Faculty of Medical Sciences Postdoc Symposium, Newcastle University, 2024
- David Blow Poster Prize, Biological Structure Group of the British Crystallographic Association, 2024
- Presentation Prize, Molecular Mechanisms of Life Theme Summer Event, Newcastle University, 2022
- David Thompson Scholarship for outstanding Tripos results, Homerton College, Cambridge 2015

Invited talks

- Microbiology and Systems Biology Guest Seminar, Department of Biochemistry, University of Oxford, UK (2024)
- NUBI Live!, Newcastle upon Tyne, UK (2022)

Conference attendance and presentations

- (poster) Bacterial Cell Surfaces, Gordon Research Conference/Seminar, Maine, USA (2024)
- (talk, poster) British Crystallographic Association Spring Meeting, Leeds, UK (2024)
- (talk) FEMS Congress of Baltic Microbiologists, Vilnius, Lithuania (2023)
- (poster) Bacterial Cell Surfaces, Gordon Research Conference, Vermont, USA (2022)
- (talk) Bacterial Cell Surfaces, Gordon Research Seminar, Vermont, USA (2022)
- (talk) 6th Midlands Molecular Microbiology Meeting, Nottingham, UK (2019)
- (poster) Annual Biochemistry Graduate Symposium, Oxford, UK (2019)
- 5th Midlands Molecular Microbiology Meeting, Warwick, UK (2018)
- 5th Molecular Microbiology and 4th Midlands Molecular Microbiology Meeting, Birmingham, UK (2017)
- 19th IUPA Congress and 11th EBSA Congress, Edinburgh, UK (2017)
- (poster) Medical Sciences Doctoral Training Centre Annual Symposium, Oxford, UK (2017)
- 3rd Midlands Molecular Microbiology Meeting, Leicester, UK (2017)

Academic service

- Scientific Committee Member, CCP4 Northern Structural Biology Meeting, Newcastle, UK (2024).
- Reviewer for *Nature Communications* and *Communications Biology* (2023-present).
- Chairing: Newcastle University Early Career Researcher Celebration (2023), NUBI Live! (2023).
- Member of Newcastle University Biosciences Institute Research Associate Committee (2022-present).